



Department of Computer Systems Engineering,
University of Engineering and Technology, Peshawar,
Pakistan

Midterm Exam (Spring 2023)

Time: 1 Hour 45 Minutes

Paper: CSE-403 Database Management System

Marks: 80

Note: Attempt all questions on answer sheet.

Paper Part B

Q II. Draw an E-R Diagram for each of the following scenario. Ensure that all entities, attributes, relationships, degree of relationships, types of relationship, cardinalities and the given business rules are correctly depicted with appropriate notations in the ERD. If you believe that you need to make additional assumptions, clearly state them for each situation. (Marks=70) (CLO-2)

a) An Employee may be assigned a company CAR, but some Employees do not qualify for a company CAR. A CAR will be assigned to one and only one Employee. Each car has associated with it zero to any number of recorded accidents.

b) An animal may eat many other different animals or may not eat any animals (vegetarian). An animal may be eaten by many different animals or may not be eaten by any animal at all.

c) In a Manufacturing Company, a part has at least one alternative part (or substitute part). Some parts do not serve as an alternative (substitute) for any parts. Also, a part may serve as an alternative for many different parts.

d) Create an ERD for a service station business that provides goods and services to its customers. Typically a customer comes in with their vehicle and requests certain work to be performed. For example a customer may request an oil change and for a new set of four tires to be provided and installed. The work items that can be performed or supplied can be of two types: a service (such as the oil change) and actual physical items (such as litres of oil). There will be several services that can be performed such as tire installation, changing oil, or fixing a flat tire. Each of these will have some cost to be charged to a customer. There are many concrete items that are supplied and charged to a customer such as fan belts, litres of oil, or tires – these are things that are kept in inventory. Consider creating a hierarchy for products (goods / services); make up reasonable attributes.

This service station has customers that fall into two groups: some are private individuals and others are businesses. Individuals will have a first name, last name, address and phone number. A business will have a business name, address, phone number and a contact person who has a first name and last name. Consider creating a hierarchy for customers.

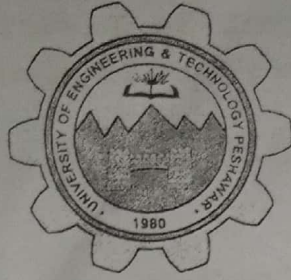
The service station needs to keep track of all the goods and services it provides to its customers so that it has a historical record and knows what it has charged to each customer. Each visit to the service station by a customer will generate a work order that keeps track of the work that was done for the customer's vehicle. Vehicles have license plate numbers, and other attributes to describe them (make, model, colour, ...). For each visit of a customer to the station the system needs to know the date the visit occurred, the details of the work performed and goods provided, and the total charge to the customer.

f) An institute facilitates its students to participate in three types of sports events: Long Jump, Discus Throw and 100- Meter race. The following attributes are recorded for each event:

- Long Jump: Student Roll Number, Name, House, Age, Recorded Jump
- Discus Throw: Student Roll Number, Name, House, Age, Distance covered
- 100 m. Race: Student Roll Number, Name, House, Age, Time taken

Apply rule of generalization and develop an EER model segment using Visio notation. Assume that each of this sport event can be a part of exactly one of these subtypes.

Q III. List 5 potential benefits and costs/risks associated with Database approach over conventional file systems. (Marks=10)



Department of Computer Systems Engineering
University of Engineering & Technology Peshawar
6th Semester Final-term Examination, Spring 2023

Course Title: **Technical Writing**

Course Code: CSE311

Total Marks: 50

Maximum Weightage: 50%

Time allowed: 02 Hours

Name: Farrukh Adeel

Reg #: 20pucse1883

- Attempt all five questions; all of them carrying equal marks.
- Do not write anything on this question paper except your name and registration number.
- Mobile Phones, Smart Watches and all types of Programmable Devices are strictly prohibited.

Q1 – You are employed as a Network & Security Officer in a large multinational company. Your company's wireless security system has been facing problems due to its extensive use or may be due to some other causes. Write a letter to the purchase manager, explaining the problem and requesting the purchase of a new multi-function device. Provide the rationale/justification for the expense.

Affective-2 (Responding)

Q2 – In your candid opinion, how should the content, delivery and assessment of this course "Technical Writing" be changed for the upcoming batches?

Affective-3 (Valuing)

Q3 – What are the essential ingredients of a good proposal? Explain/Justify with the help of your own FYP proposal that you wrote as an assignment for this course.

Affective-4 (Organization)

Q4 –

Congratulation! You've completed your degree and got a terrific job in your field. One of your instructors was especially helpful to you when you were a student. Besides other facilitations, this instructor also wrote an effective letter of recommendation that was instrumental in helping you obtain your job. Now write an email of thanks/appreciation to your instructor.

Q5 –

a) Following are the 7Cs of Effective Business Communication. Write a line about each.

1. Correctness
2. Clarity
3. Conciseness
4. Completeness
5. Consideration
6. Concreteness
7. Courtesy

b) What do you understand by the phrase "Tell them what you are going to tell them then tell them and then tell them what you have told them"?

Name: Farrukh Adeel
Registration: 20pwcse1883

Department of Computer Systems Engineering
University of Engineering & Technology Peshawar

Digital System Design
CSE 308

Midterm Examination Spring 2023
13 April 2023, Duration: 120 Minutes

****Exam Rules****

Please read carefully before proceeding.

- 1- This exam is closed books/notes/Internet.
- 2- Answer all problems on the answer sheet.
- 3- Problems will not be interpreted during the exam.

Good Luck!

Problem 1. (25 pts.)

Below is an RTL (or Dataflow) description for a circuit.

```
module RTL_circuit (x, y, a, b, c);  
    input a, b, c;  
    output x, y;  
    wire a, b, c, x, y;  
    wire na, nb, nc, t3, t4, t5;  
    assign na = !a;  
    assign nb = !b;  
    assign nc = !c;  
    assign t3 = na && b && c;  
    assign t4 = a && nb && c;  
    assign t5 = a && b && nc;  
    assign x = t3 || t5;  
    assign y = a || t4;  
endmodule
```

- 1(a) (3 pts.) Give a Verilog statement that instantiates the above RTL_circuit, with the instance name MID. When you instantiate the circuit, use the same names for wires as is used in the module port list.

RTL_circuit MID(x, y, a, b, c);

- 1(b) (6 pts.) Rewrite the RTL_circuit using Verilog built-in primitives and structural Verilog. Part of the module is done for you.

```
module struct_circuit (x, y, a, b, c);  
    input a, b, c;  
    output x, y;  
    //Write your code here  
    wire na, nb, nc, t3, t4, t5;  
endmodule
```

*not n1(na, a);
not n2(nb, b);
not n3(nc, c);
and ta(t3, na, b, c);
and tb(t4, a, nb, c);
and tc(t5, a, b, nc);
or o1(x, t3, t5);
or o2(y, a, t4);*

- 1(c) (3 pts.) Draw a gate-level diagram for your module in 1(b). Label all nets on the diagram.

- 1(d) (6 pts.) Rewrite the RTL_circuit using behavioral Verilog. Part of the module is done for you.

```

module behav_circuit (x, y, a, b, c);
    input a, b, c;
    output x, y;
    //Write your code here

```

```

endmodule

```

1(e) (7 pts.) Write the output of RTL_circuit for the following test bench.

```

//test bench for RTL_circuit
module test_RTL_circuit;

    reg a, b, c;
    wire x, y;

    RTL_circuit RTLC (x, y, a, b, c);

    Initial begin
        a = 1; b = 1; c = 1;
        #30 a = 0; b = 1; c = 1;
    end

    initial begin
        0 $display ($time, " x = %b and y = %b", x, y);
        6 #6 $display ($time, " x = %b and y = %b", x, y);
        11 #5 $display ($time, " x = %b and y = %b", x, y);
        31 #20 $display ($time, " x = %b and y = %b", x, y);
        36 #5 $display ($time, " x = %b and y = %b", x, y);
        42 #6 $display ($time, " x = %b and y = %b", x, y);
        48 #8 $display ($time, " x = %b and y = %b", x, y);
    end

end

endmodule

```

$t = 0 =$

$t_3 = 0 \quad t_4 = 0, t_5 = 0 \quad (0, 0)$

$t = 6; \quad t_3 = 0, t_4 = 0, t_5 = 0 \quad (0, 0)$

$t = 11; \quad t_3 = 0, t_4 = 0, t_5 = 0 \quad ($

Problem 2. (10 pts.)

In this problem, design a 3-bit counter (the circuit diagram and state table are shown in Figure 1) using as building block the D flip-flop given in Figure 2.

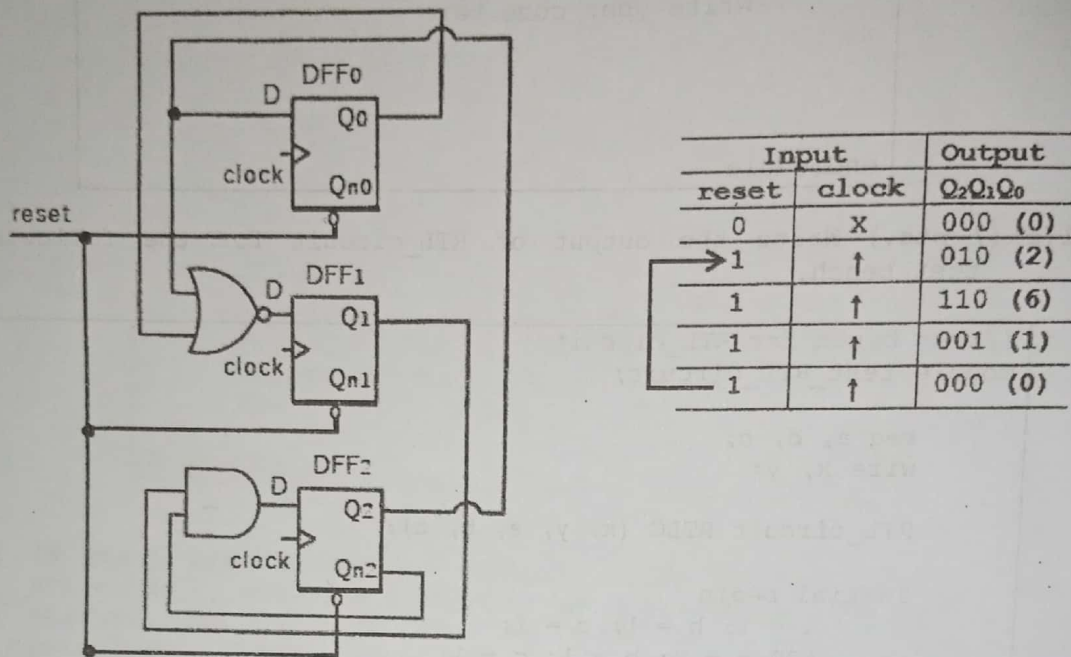


Figure 1. The circuit diagram (left) and state table (right) of the 3-bit counter

```

module DFF (D, clock, reset, Q, Qn);
    input D; // Data input
    input clock; // Clock input
    input reset; // Asynchronous active-low reset
    output Q, Qn; // Outputs Q and Q'
    reg Q;

    always @ (posedge clock or negedge reset)
        if (reset==1'b0)
            Q <= 0;
        else
            Q <= D;

    assign Qn = ~Q;
endmodule

```

Figure 2. Verilog code for rising-edge D flip-flop with asynchronous active-low reset

module TOP;

input clk, reset, [7:0] D

output [2:0] out;

Counter (clock, reset, out[2:0])

max ([2:0] D, Sel(out[2:0]), 1[7:0])

The suggested skeleton file for the counter has been written below. The module has 2 inputs - **clock** and **reset** which is active-low. The output is **out** which is 3-bit in size.

```
module counter (clock, reset, out);
```

```
    input clock, reset;
```

```
    output [2:0] out;
```

```
    // Write your code here
```

```
    Reg [2:0] out wire [2:0] f;
```

```
    DFF D0, C1, D0, clock, reset, f[0], ~out[2:0]
```

```
endmodule
```

Problem 3. (10 pts.)

In this problem, write Verilog code for an 8-to-1 multiplexer. In this case, the value on the 3-bit select line will route 1 of 8 inputs to the output. This module is purely combinatorial.

The following are the ports of the module:

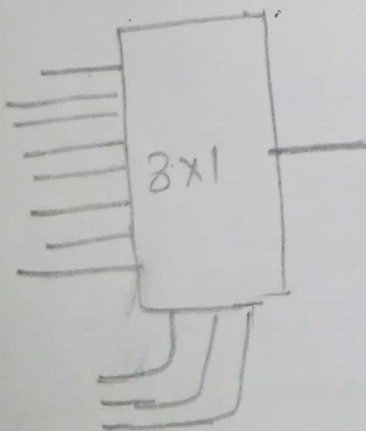
Sel	3-bit select line
I0, I1, I2, I3, I4, I5, I6, and I7	1-bit data inputs
OUT	1-bit output

Problem 4. (10 pts.)

For this design, combine the counter (from **Problem 2**) with the multiplexer (from **Problem 3**) to create a circuit such that the output of the counter controls the select lines of the multiplexer.

This top-level design has the following port definitions:

clk	1-bit clock
reset	1-bit reset line
D0, D1, D2, D3, D4, D5, D6, and D7	1-bit data inputs
OUT	1-bit output



```
module (Out, Sel [2:0], I [7:0])
```

```
    Input [2:0] S
```

```
    Input [7:0] I;
```

```
    output I;
```

```
    @always (I, Sel) begin
```

```
        if (S)
```

```
            case (S)
                0: I <= I[0];
```

```
                1: I <= I[1];
```

```
            endcase
        end
```



Department of Computer Systems Engineering
University of Engineering & Technology
Peshawar

Microprocessor Based System Design (MBSD)
6th Semester Mid-Term, Spring 2023

Max. Time: 2 hours

Max. Points: 20

Instructions:

1. Attempt ALL questions.
2. Exam is open book and open notes.
3. Cell phones and Laptops are strictly prohibited.
4. Exam is worth 20% of the final grade.

Q 1).

8-points

CLO-5/PLO-3 [Cognitive Domain: Synthesis]

Design a system, where the Software in C will generate, a signal of

- A. 0.25KHz with a duty cycle of 10% on P2.0 pin.
 - B. Whenever a user presses a button at (P3.2), the signal toggles to 0.5KHz with a duty cycle of 50%.
 - C. Again, pressing the same button will generate a signal of 1KHz with a duty cycle of 75%.
A third time button press will result in the generation of case A and so on.
- Draw the schematic diagram showing clearly the button circuit and oscilloscope.
 - Draw the timing diagram with cursors clearly showing the time period with appropriate units.
 - Assuming oscillator clock of 24MHz is used.
 - Use **timer** interrupt. Use External interrupts.
 - Polling is not allowed.

Name & Reg No: _____

ame & Reg

Q3).

Trans

Q2).

8-points

CLO-3/PLO-2 [Cognitive Domain: Application]

Fill the table by solving the values of variables in the program.

```
unsigned char x = 0x12, y=0, z = 0, R = 7 ;  
void main(void)
```

```
{  
  
    while(y<5) {  
        x = 0x80 - y;  
        x^= x;  
        x += 0x7F;  
        x = x>>y;  
        R += (R << 1);  
        z += (x+y+R);  
  
        y++;  
    }  
}
```

	X	R	Z
Y=0	7F	15	04
Y=1	3E	3E	13
Y=2	1D	BD	F1
Y=3	0C	37	3A
Y=4	07	A5	29

main: Application
8-points
R = 7

me & Reg No: _____

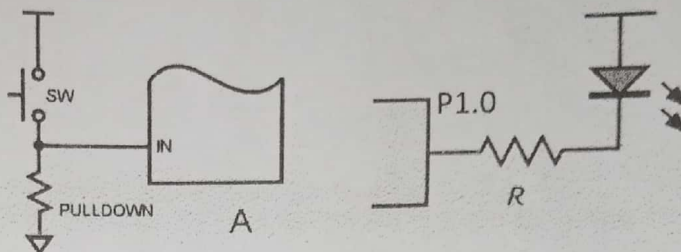
Q 3).

4-points

CLO-2/PLO-1 [Cognitive Domain: Comprehension]

Translate the following tasks into C code.

- i) If we have an active-high button (A) at P2.5 pin and an *active-low* LED at P1.0 as shown below,



P2.5 = 0X04; led = 1;

Scan the button using polling

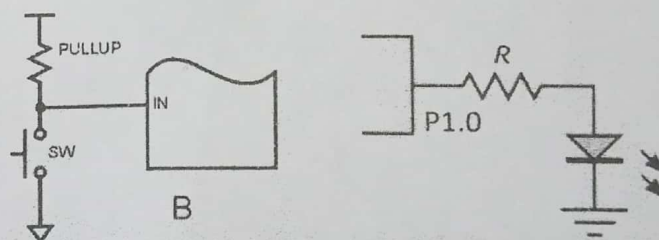
While (P2.5 == 1)

{

P1.0 = 0 // Button pressed, TURN ON the active-low LED at P1.0

}

- ii) If we have an active-low button (B) at P2.5 and an *active-high* LED at P1.0 as shown below,



P2.5 = 0X74; led = 0;

Scan the button, using polling

While (P2.5 == 0)

{

P1.0 = 1 // Button pressed, TURN ON the active-high LED at P1.0

}

Name & Reg No: Farrukh Adeel 1883



Department of Computer Systems Engineering
University of Engineering & Technology
Peshawar

Microprocessor Based System Design (MBSD)
6th Semester Final-Term, Spring 2023

Max. Time: 2 hours

Max. Points: 50

Instructions:

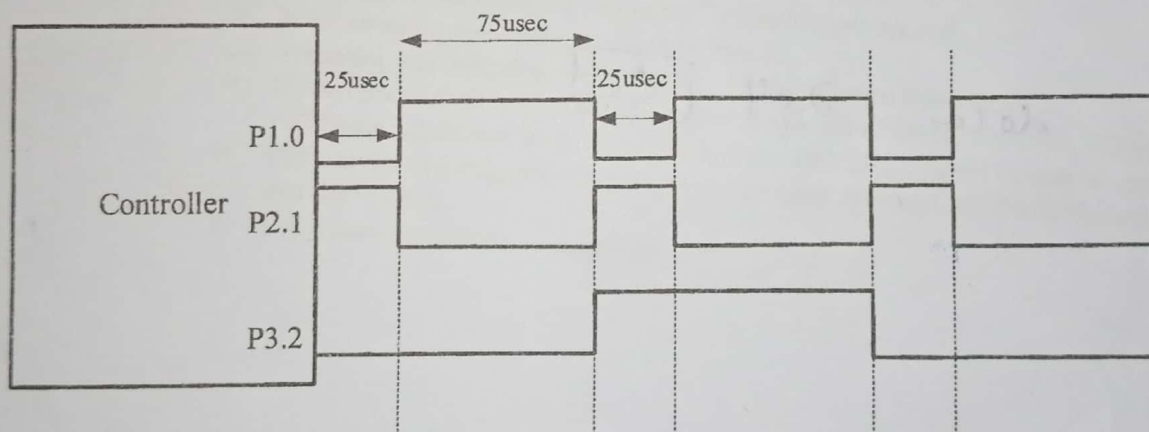
1. Attempt ALL questions.
2. Exam is open book and open notes.
3. Exam is worth 50% of the final grade.

Q 1).

15-points

Design CLO-2/PLO-1 [Cognitive Domain: Comprehension]

- A. Translate the timing diagram below and write the code for it to generate these periodic signals. Use timer interrupts for this purpose.



Can you generate all three signals using a single timer?

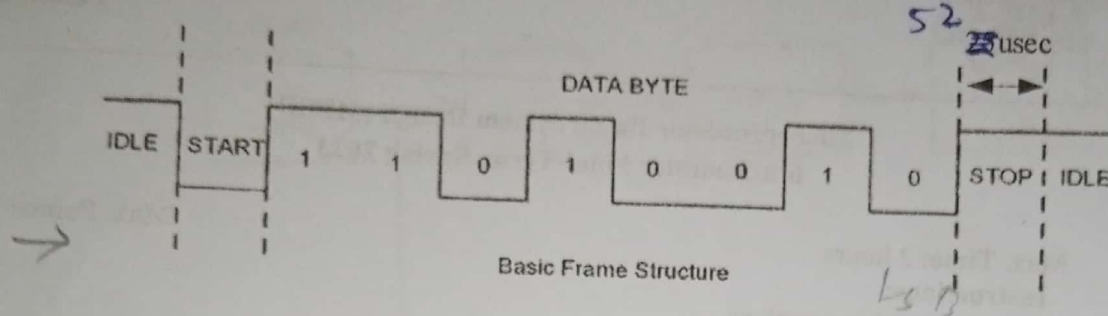
Name & Reg No: 20pwec1883

Q 2).

15-points

CLO-3/PLO-3 [Cognitive Domain: Application]

An Engineer at NXP semiconductor found this data on oscilloscope when he probed the UART communication between two 8051 controllers.



Write the C code for it by Solving the bit-duration of the above timing diagram.
What data will be received by the receiver?
What is the transmission rate?

01001011

20 / 1. overhang

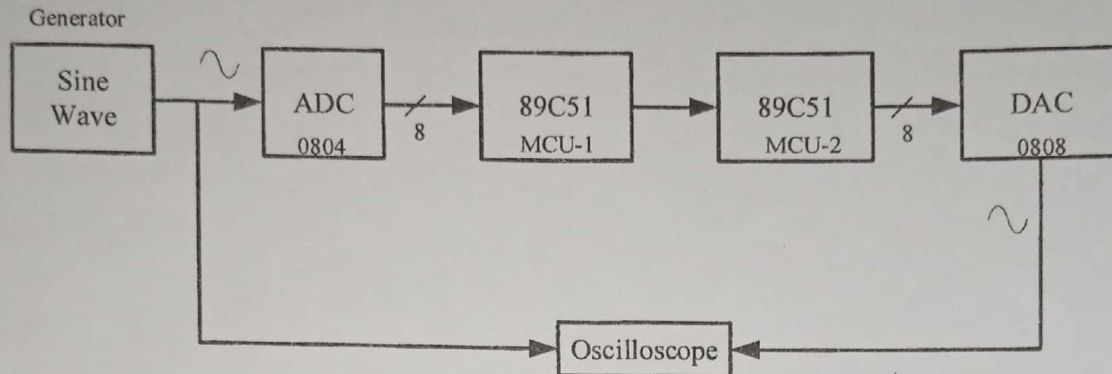
delay 204 CC

Q 3.

20-points

CLO-5/PLO-3 [Cognitive Domain: Synthesis]

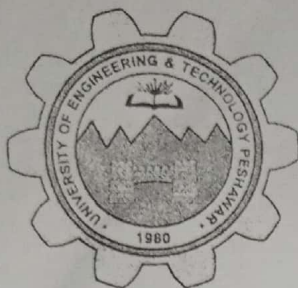
Design a system, where the Software in C is used to interface an ADC and DAC, to two 89C51 microcontrollers as shown below in figure,



Develop the system as shown above.

The oscillator frequency of both microcontrollers is fixed at **22.118MHz**. Keeping in view the fastest possible transmission rate of serial comm and ADC conversion rate. How much the frequency of input signal can be increased? Run the system at that frequency and write the code for it.

- What will be transmission rate of MCU-1 in bits per second.
- What sampling rate will you choose for ADC.
- Assuming sampling frequency = $(10 \times \text{Input sine wave frequency})$
- If you choose X Hz as the fastest possible frequency of input sine wave. Which component will be the bottleneck at $(X+1)$ Hz. Will it be ADC, DAC, Serial Comm or something else?
- Draw the timing diagram of input sine wave signal and output of DAC. Show the amplitude and time period.
- Use timers for precise calculations.



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6th Semester Final-term Examination, Spring 2023

Course Title: **Technical Writing**

Course Code: CSE311

Total Marks: 50

Maximum Weightage: 50%

Time allowed: 02 Hours

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Affective-2 (Responding)

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Affective-3 (Valuing)

Q3 – What are the essential ingredients of a good proposal? Explain/Justify with the help of your own FYP proposal that you wrote as an assignment for this course.

Affective-4 (Organization)

Q4 –

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Q5 –

a) Following are the 7Cs of Effective Business Communication. Write a line about each.

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2. Clarity
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4. Completeness
5. Consideration
6. Concreteness
7. Courtesy

b) What do you understand by the phrase "Tell them what you are going to tell them then tell them and then tell them what you have told them"?